

Variance Component Modelling of Nuclear Family Data: From Trio Designs to Full ACDE Estimation in R

Contents

1 Introduction

Biometrical genetics aims to partition phenotypic variation into genetic and environmental components:

- Additive genetic effects (A)
- Dominance genetic effects (D)
- Shared environmental effects (C)
- Unique environmental effects (E)

Twin studies traditionally estimate ACE or ADE models. Modern population cohorts, however, frequently contain nuclear family data. This paper presents a unified framework showing:

- what can be estimated from trio data,
- why siblings solve identifiability,
- how to estimate AE/ACE/ADE/ACDE using linear mixed models in R.

2 General Variance Decomposition

For phenotype y :

$$y = \mu + a + d + c + e$$

$$Var(y) = \sigma_A^2 + \sigma_D^2 + \sigma_C^2 + \sigma_E^2$$

3 General Pedigree Covariance

For relatives i, k :

$$Cov(y_i, y_k) = 2\phi_{ik}\sigma_A^2 + \Delta_{ik}\sigma_D^2 + \gamma_{ik}\sigma_C^2$$

- ϕ_{ik} : kinship coefficient
- Δ_{ik} : probability of sharing two alleles IBD
- γ_{ik} : shared family environment indicator

4 Nuclear Family Trio Design

Consider mother–father–child trios.

4.1 Expected Covariances

Pair	2ϕ	Δ	γ
Mother–Father	0	0	1
Mother–Child	$\frac{1}{2}$	0	1
Father–Child	$\frac{1}{2}$	0	1

4.2 Key Consequence

Dominance deviations are not transmitted:

$$\Delta_{ik} = 0 \quad \text{for all trio pairs}$$

Thus dominance contributes only to individual variance:

$$Var(y) = \sigma_A^2 + \sigma_D^2 + \sigma_E^2$$

4.3 Identifiable Family Resemblance

Trio data contain only one familial covariance:

$$Cov_{family} = \frac{1}{2}\sigma_A^2 + \sigma_F^2$$

$$\sigma_F^2 = \begin{cases} \sigma_C^2 & \text{ACE interpretation} \\ \text{residual familial resemblance} & \text{ADE interpretation} \end{cases}$$

Hence:

Trio data allow AE, ACE or ADE — but not ACDE.

5 Why Adding Siblings Solves Identifiability

Sibling covariance:

$$Cov(\text{sibling}_1, \text{sibling}_2) = \frac{1}{2}A + C + \frac{1}{4}D$$

Design	Identifiable components
Trio	AE, ACE or ADE
Family with siblings	ACDE

6 ACDE Covariance Matrix

For parents (P_f, P_m) and siblings (S_1, S_2) :

$$\Sigma = \begin{pmatrix} V_P & V_C & \frac{1}{2}V_A + V_C & \frac{1}{2}V_A + V_C \\ V_C & V_P & \frac{1}{2}V_A + V_C & \frac{1}{2}V_A + V_C \\ \frac{1}{2}V_A + V_C & \frac{1}{2}V_A + V_C & V_P & \frac{1}{2}V_A + V_C + \frac{1}{4}V_D \\ \frac{1}{2}V_A + V_C & \frac{1}{2}V_A + V_C & \frac{1}{2}V_A + V_C + \frac{1}{4}V_D & V_P \end{pmatrix}$$

7 Likelihood Formulation

$$\mathbf{y}_i \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

$$\log f(\mathbf{y}_i) = -\frac{1}{2} [\log |\Sigma| + \mathbf{y}_i^\top \Sigma^{-1} \mathbf{y}_i + k \log(2\pi)]$$

Variance parameterisation:

$$V_A = e^{\theta_A}, \quad V_C = e^{\theta_C}, \quad V_D = e^{\theta_D}, \quad V_E = e^{\theta_E}$$

8 Linear Mixed Model Representation

Stacked observations:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}_A\mathbf{a} + \mathbf{Z}_F\mathbf{f} + \boldsymbol{\varepsilon}$$

$$\mathbf{a} \sim N(0, \sigma_A^2 \mathbf{A}) \quad (1)$$

$$\mathbf{f} \sim N(0, \sigma_F^2 \mathbf{I}) \quad (2)$$

$$\boldsymbol{\varepsilon} \sim N(0, \sigma_E^2 \mathbf{I}) \quad (3)$$

9 Data Restructuring in R

```
library(nlme)
```

```
longdata <- reshape(
  widelong,
  varying=c("bmi1", "bmi2", "bmi3"),
  v.names="bmi",
  timevar="member",
  times=c("mother", "father", "child"),
  idvar="famid",
  direction="long"
)

longdata$Acoef <- ifelse(longdata$member=="child", 1, 0.5)
longdata$Ccoef <- 1
longdata$Dcoef <- 1
```

10 AE Model

$$Var(y) = \sigma_A^2 + \sigma_E^2$$

```

model_AE <- lme(
  bmi~1,
  random=list(
    famid=pdDiag(~Acoef-1),
    indid=pdDiag(~Dcoef-1)
  ),
  data=longdata, method="REML")

```

11 ACE Model

$$Var(y) = \sigma_A^2 + \sigma_C^2 + \sigma_E^2$$

```

model_ACE <- lme(
  bmi~1,
  random=list(
    famid=pdDiag(~Acoef+Ccoef-1),
    indid=pdDiag(~Dcoef-1)
  ),
  data=longdata, method="REML")

```

12 ADE Model

$$Var(y) = \sigma_A^2 + \sigma_D^2 + \sigma_E^2$$

```

model_ADE <- lme(
  bmi~1,
  random=list(
    famid=pdDiag(~Acoef-1),
    sibshipid=pdDiag(~Dcoef-1)
  )

```

```
),  
data=longdata, method="REML")
```

13 Extracting Variance Components

```
VarCorr(model_AE)  
VarCorr(model_ACE)  
VarCorr(model_ADE)
```

14 Model Comparison

```
anova(model_AE, model_ACE)  
anova(model_AE, model_ADE)
```

15 Derived Quantities

$$h^2 = \frac{V_A}{V_P}, \quad c^2 = \frac{V_C}{V_P}, \quad d^2 = \frac{V_D}{V_P}, \quad e^2 = \frac{V_E}{V_P}$$

16 Conclusion

- Trio data estimate additive genetic variance and one familial component.
- Siblings enable separation of shared environment and dominance.
- Linear mixed models provide a practical implementation in R.